



Round 2 Section 2 – Case Study Information Pack

Section 2: Case Study – Winning Ways

Relates to Questions 11-23

30 Marks available in this Section

INTRODUCTION

On 31 December 2018 you are to take over ownership of a sporting franchise. Currently you are looking at what strategy you want to take with the team over the next 3 years. As part of this you want to evaluate the impact on the cash balance of these strategies. Your team currently has one superstar player. However, your team only finished fifth last season. The scenarios you are looking into are:

Scenario 1) Keep the same players as last season for likely similar results (ie “as is”)

Scenario 2) Trade away superstar player for future draft picks (ie “tank”)

Scenario 3) Sign another superstar player and other veterans (ie “super-team”)

You are required to build a financial model to determine the cash balance at the end of 3 years under each of the scenarios.

The competition runs from 1 April to 31 October with 8 games held every month (50% at home). The salaries paid are all aligned to calendar years. You are interested in the period from 1 January 2019 to 31 December 2021. The number of wins expected in each month under each scenario is included in the attached spreadsheet.

All cash-flows are received/paid on the last day of the month. No tax is paid (other than the luxury tax).

REVENUES

Gate receipts are the biggest stream of revenue for the team. Percentage attendance at each game is calculated as:

$$\% \text{ Attendance} = (40\% + \text{Min} \left(40\%, \frac{\text{wins in previous two calendar months}}{\text{ticket price}} * 0.8 \right) + 20\% * C)$$

‘Wins in previous two calendar months’ equals the total wins in the two complete months prior to the current one. In April and May of each season this is treated as 10.

C is a factor that measures whether the team is likely to finish in the top 4 and is 1 if the team is still in contention, and 0 if they are deemed to not be able to make the top 4. This is determined by whether on the first of the month, winning 80% of the remaining games for the season will give them a top 4 finish.

The stadium holds 10,000 seats. Ticket prices are locked in for the 3 year period and you have decided to vary them based on the scenario:

	Scenario 1	Scenario 2	Scenario 3
Average ticket price	\$28	\$20	\$36

100% of the ticket revenue is received for all home games, with no amount received for away games.



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TV rights have been sold by the league and this results in \$150,000 being paid to the team in every month over the 3 year period (not just when games are played). The team also receives sponsorship income based on the number of wins in the previous season. This is based on the formula:

$$\text{Annual sponsorship} = \text{Min} \left(100\%, \frac{\text{Total wins in previous season}}{50} \right) * \$3.4m$$

Sponsorship is paid monthly, spread evenly over the year. The team won 34 games in 2018.

If the team finishes in the top four, it receives prize money. This is received as a lump sum on 30 November immediately after the relevant season. Based on recent history the number of wins required for each place is reasonably well known as follows:

	First	Second	Third	Fourth
Number of wins required	50	45	40	35
Prize money	\$6m	\$4.5m	\$3m	\$2.5m



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EXPENSES

Operating expenses are \$200,000 per month, and stadium fees are \$50,000 per home game. Expected annual salaries for each year and scenario are:

Calendar year	Scenario 1	Scenario 2	Scenario 3
2019	\$4,563,760	\$2,551,833	\$6,635,980
2020	\$4,621,084	\$2,646,091	\$7,433,709
2021	\$4,674,029	\$2,721,511	\$7,268,784

Salaries are paid evenly throughout the year and all players are assumed to remain on the roster for the complete year.

Salaries are subject to a soft salary cap, where any amounts over the salary cap (\$5m) have an additional luxury tax levied on them. The tax amounts are as follows:

In calendar year	Tax	Tax charged on
Salaries <= 5,000,000	0%	
5,000,000< Salaries <= 6,000,000	100%	Only amounts between \$5m and \$6m
6,000,000< Salaries <= 7,000,000	200%	Only amounts between \$6m and \$7m
Salaries > 7,000,000	300%	Only amounts greater than \$7m

Luxury tax is paid at the end of each month based on the cumulative salary costs for the year thus far. The table below shows an example of how this works:

	30 September	31 October	30 November	31 December
Cumulative salaries in year	\$4,900,000	\$5,700,000	\$6,300,000	\$7,100,000
Charged at 100%	\$0	\$700,000	\$300,000	\$0
Charged at 200%	\$0	\$0	\$300,000	\$700,000
Charged at 300%	\$0	\$0	\$0	\$100,000
Tax paid	\$0	\$700,000	\$900,000	\$1,700,000

Note that only Scenario 3 has a salary bill high enough to trigger luxury tax.



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DEBT

To purchase the team, you borrowed \$20m. This debt has interest paid monthly at 5% pa. Two repayments of \$500,000 are scheduled to be made annually. These are paid on 31 March and 30 September each year. No additional repayments can be made, or further debt drawn down.

You also have the ability to access a revolving credit line. This has interest paid monthly at 7.5% pa.

For both interest calculations, the monthly interest rate should be calculated as:

$$(1 + \text{Annual Interest Rate})^{\frac{1}{12}} - 1$$

The revolving credit line is used to ensure that in each month the cash balance is greater than the minimum cash balance. Any cash amounts above the minimum cash balance can be used to pay back the outstanding revolving credit. There are no fees associated with using the revolving credit.

OTHER

You currently have \$2m of cash on hand. A minimum cash balance of \$200,000 must be held at the end of each month.

This information is relevant for questions 21 and 22 only

The owners of the stadium have offered to add an additional 2,000 seats but will increase the stadium fees by 50% per game. You now expect attendance to follow this formula:

$$\% \text{ Attendance} = 33\% + \text{Min} \left(47\%, \frac{\text{wins in previous two calendar months}}{\text{ticket price}} * 0.8 \right) + 20\% * C$$

Prepare your model and then use it to answer questions 11-22.

When finished, please upload your workbook (Question 23).



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QUESTIONS

Question 11

In Scenario 1, what is the first month they fall out of top 4 contention (ie $C = 0$)? [2 marks]

- A: April 2019
- B: May 2019
- C: June 2019
- D: July 2019
- E: August 2019
- F: September 2019
- G: October 2019
- H: November 2019
- I: December 2019

Question 12

In Scenario 1, what is the total number of tickets sold in the month of September 2019? [2 marks]

- A: 20,570
- B: 20,571
- C: 20,572
- D: 20,573
- E: 20,574
- F: 20,575
- G: 20,576
- H: 20,577
- I: 20,578



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Question 13

In Scenario 1, what is the total ticket revenue in 2020? [2 marks]

- A: 6,820,000
- B: 6,830,000
- C: 6,840,000
- D: 6,850,000
- E: 6,860,000
- F: 6,870,000
- G: 6,880,000
- H: 6,890,000
- I: 6,900,000

Question 14

In Scenario 1, what is the total revenue (including prize money) over the 3 year period? [3 marks]

- A: 34,070,000
- B: 34,071,000
- C: 34,072,000
- D: 34,073,000
- E: 34,074,000
- F: 34,075,000
- G: 34,076,000
- H: 34,077,000
- I: 34,078,000



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Question 15

In Scenario 1, what is the total amount of revenue less expenses excluding interest over the 3 year period? [2 marks]

- A: 8,817,125
- B: 8,817,126
- C: 8,817,127
- D: 8,817,128
- E: 8,817,129
- F: 8,817,130
- G: 8,817,131
- H: 8,817,132
- I: 8,817,133

Question 16

In Scenario 1, what is the cash balance at 31 December 2021? [4 marks]

- A: 5,102,705
- B: 5,102,706
- C: 5,102,707
- D: 5,102,708
- E: 5,102,709
- F: 5,102,710
- G: 5,102,711
- H: 5,102,712
- I: 5,102,713



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Question 17

In Scenario 2, what is the total interest paid on the revolving debt over the 3 year period? [3 marks]

- A: 21,395
- B: 21,396
- C: 21,397
- D: 21,398
- E: 21,399
- F: 21,400
- G: 21,401
- H: 21,402
- I: 21,403

Question 18

In Scenario 2, what is the cash balance at 31 December 2021? [3 marks]

- A: 2,965,793
- B: 2,965,794
- C: 2,965,795
- D: 2,965,796
- E: 2,965,797
- F: 2,965,798
- G: 2,965,799
- H: 2,965,800
- I: 2,965,801



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Question 19

In Scenario 3, what are the total expenses paid on 31 December 2021? [2 marks]

A: 2,285,980

B: 2,285,981

C: 2,285,982

D: 2,285,983

E: 2,285,984

F: 2,285,985

G: 2,285,986

H: 2,285,987

I: 2,285,988

Question 20

**In Scenario 3, what is the cash balance on 31 December 2021? Note: Enter to the nearest dollar.
[3 marks]**



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This information is relevant for questions 21 and 22 only

The owners of the stadium have offered to add an additional 2,000 seats but will increase the stadium fees by 50% per game. You now expect attendance to follow this formula:

$$\text{Attendance} = 33\% + \text{Min}\left(47\%, \frac{\text{wins in previous two calendar months}}{\text{ticket price}} * 0.8\right) + 20\% * C$$

Question 21

Under which scenario(s) does this offer result in a higher cash balance on 31 December 2021?
[1 mark]

- A: None
- B: Scenario 1
- C: Scenario 2
- D: Scenario 3
- E: Scenario 1&3
- F: Scenario 1&2
- G: Scenario 2&3
- H: All scenarios

Question 22

If you elect to take up this offer, what is the sum across all three scenarios of the increase in cash balance at 31 December 2021? Note: Enter to the nearest dollar, if it's a decrease then enter a negative number. [3 marks]

Question 23

Please upload your workbook for this section.